



PRE-BOOTCAMP READINESS SESSION

AI in Biomed Boot Camp

Getting everyone set up on AWS — GW Research Gateway

60 minutes · Rocky Linux 8 Remote Desktop · Jupyter + Ollama

Intro

Getting Up and Running

Get help offline by emailing:
RTSHelp@gwu.edu

Screenshots, setup guide, and notebooks
are in the BootcampSquared GitHub repo.
Scan the QR code to open it.



What is Research Gateway?

Pegasus and Research Gateway

- Pegasus is GW's flagship supercomputer that features state of the art GPUs and compute capability. However, storage is unavailable for this bootcamp readiness window.
- Research Gateway is our portal to the AWS Cloud and gives everyone a working Linux desktop environment.
- That keeps the session intact and preserves the hands-on session for the Bootcamp.

What we'll cover today

- ✓ Jupyter notebooks
- ✓ PyTorch / Torch
- ✓ Ollama + local models
- ✓ bash / CLI workflows
- ✓ Python environments (venv, pip)

The main thing that does not transfer here is Slurm job scheduling.

AGENDA

What we'll cover in this hour

We start the cloud desktop first, then use the provisioning window well.

0:00–0:05

Launch the desktop

Sign in, join the project, and start the Rocky Linux desktop

0:05–0:20

GW computing + Bash basics

Research Gateway, Pegasus, cloud vs. on-prem, and the shell commands we will use today

0:20–0:35

Open Jupyter

Log into the desktop, create the venv, and launch Jupyter

0:35–0:50

Run the Ollama notebook

Open ollama_aws.ipynb, start the model download, and confirm it responds

0:50–1:00

Readiness check + help

Troubleshooting, support, and the five things that must be green

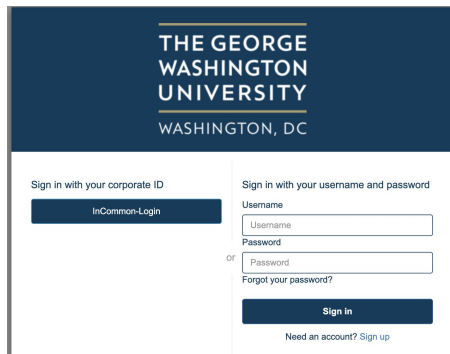
Start your cloud desktop right away

0:00–0:05

Launch first so the environment is ready by the time we reach the hands-on work.

1 Log in

rg.arc.gwu.edu → InCommon → Sign in with GW



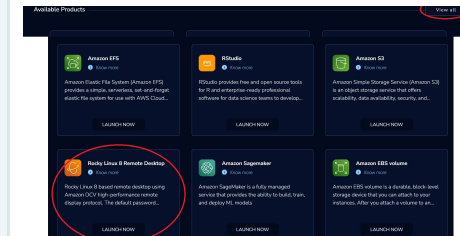
2 Open the project

Select AI_ML_Workshop



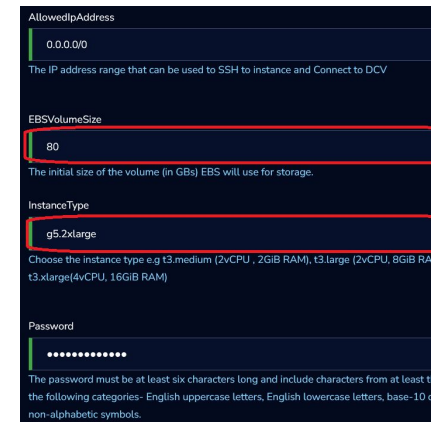
3 Choose the desktop

Rocky Linux 8 Remote Desktop



4 Configure + launch

80 GB · g5.2xlarge · Launch Now



Provisioning takes about 15 minutes, so start it before the rest of the session begins.

BEFORE THE SESSION

Complete this access check before bootcamp day

If your first login stops at "requires admin approval," get that cleared before the session starts.

1

Sign in once ahead of time

Do not make your first Research Gateway login at the start of the bootcamp.

2

Confirm the project and product

You should be able to open AI_ML_Workshop and see Rocky Linux 8 Remote Desktop.

3

Resolve blockers before the session

If approval is pending or the project is missing, contact RTShelp before bootcamp day.

What to know while the desktop provisions

0:05–0:20

1 Cloud vs. on-prem at GW

Why this bootcamp uses Research Gateway now, and when Pegasus or Raptor are the better fit for real workloads.

2 How access works

You sign in through InCommon, launch inside your project, and need to keep project budgets in mind.

3 Linux / Bash basics

Paths, files, history, tab completion, and the commands that keep people unstuck.

4 Notebooks & Google Colab

Jupyter is the working environment for this bootcamp. Colab is a useful comparison point and fallback.

Cloud and on-prem solve different problems

Cloud (Research Gateway / AWS)

- Runs as an individual desktop or VM
- Fast to provision for teaching and short projects
- Good for notebooks, package installs, and local models
- Stop the instance when you are done

On-prem (Pegasus / Raptor)

- Shared institutional compute and storage
- Better for scheduled HPC / HTC workloads
- Built around queues, allocations, and shared filesystems
- Best when you need scale, throughput, or longer-running workflows

Today we use cloud for fast, uniform setup; many production research workloads still belong on GW's on-prem systems.

Shell commands everyone should see today

Navigation and files

- `pwd` and `ls` to see where you are
- `cd` to move through directories
- `mkdir`, `cp`, `mv`, and `rm` for basic file work
- `cat` or `less` to inspect text files
- `nano` or `vi` to edit files

Habits that keep people unstuck

- Tab completion to avoid typing mistakes
- Up-arrow history to rerun and edit commands
- `man` or `--help` to check syntax quickly
- `source my_new_env/bin/activate` to enter the Python environment

The goal is not full Linux fluency; it is enough command-line comfort to navigate, activate the environment, and recover from simple mistakes.

Google Colab vs. Jupyter on your cloud desktop

Google Colab

- Runs entirely in the browser
- Easy sharing and quick experiments
- Free tiers are convenient but limited
- Sessions expire and the environment is not fully under your control

Jupyter on the cloud desktop

- You control packages, storage, and the runtime
- Persistent project workspace
- Closer to real research-system workflows
- The same environment you will use in the bootcamp

Colab is great for trying an idea; the cloud desktop is where this bootcamp actually runs.

Open Jupyter on the cloud desktop

0:20–0:35

Log into the desktop

- A new browser window opens with your desktop.
- Log in with the password you created at launch.
- If the password box is hidden, click inside the window and drag upward.

Then, in a terminal

```
git clone https://github.com/glenamac/BootcampSquared.git
sudo dnf install python3.12 -y
python3.12 -m venv my_new_env
source my_new_env/bin/activate
pip install notebook
jupyter notebook
```

Open the notebook and start the download

Once Jupyter opens, open ollama_aws.ipynb in BootcampSquared and run the model-pull cells first.

The download can run in the background while you keep moving through the notebook.

Notebook:

/BootcampSquared/ollama_aws.ipynb

How notebooks and Ollama fit together

0:35–0:50

Notebooks

- Use the notebook to run code, inspect outputs, and keep the work in one place.
- The shell still matters for setup and troubleshooting.
- This is the interface you will use throughout the bootcamp.

Ollama and models

- Ollama lets the notebook talk to a local model.
- Model choice is a tradeoff among quality, speed, and memory.
- By the end of this section, the model should be pulled and responding.

If something breaks, start here

0:50–1:00

Login fails

Check your InCommon/GW sign-in first; a first login may still need admin approval.

Instance won't launch

Check AI-ML_BioMed access, project budget, and the 80 GB / g5.2xlarge settings.

Jupyter won't start

Activate the venv first (source my_new_env/bin/activate), then try again.

Package or model missing

Re-run the notebook setup or model-pull steps and confirm the venv is active.

Where to get help

Use RTShelp or Office Hours before the bootcamp if you cannot complete the setup.

Watch the budget

Stop the instance when you are done so the shared project budget is not consumed unnecessarily.

READINESS CHECK

You're ready when...



Logged into Research Gateway (rg.arc.gwu.edu)



Launched Rocky Linux 8 Remote Desktop (80 GB · g5.2xlarge)



Jupyter running from your venv



Ran ollama_aws.ipynb end to end



Model pulled and confirmed responding in a notebook

Arrive at the bootcamp with all five green!

APPENDIX

Step-by-step setup reference

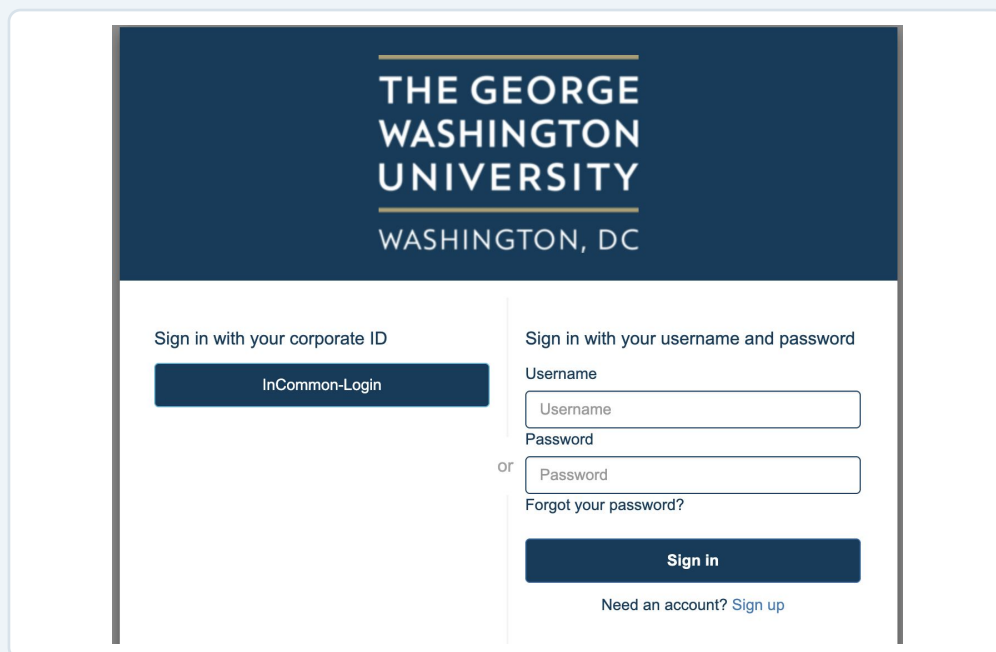
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Sign in to Research Gateway

1 Go to rg.arc.gwu.edu

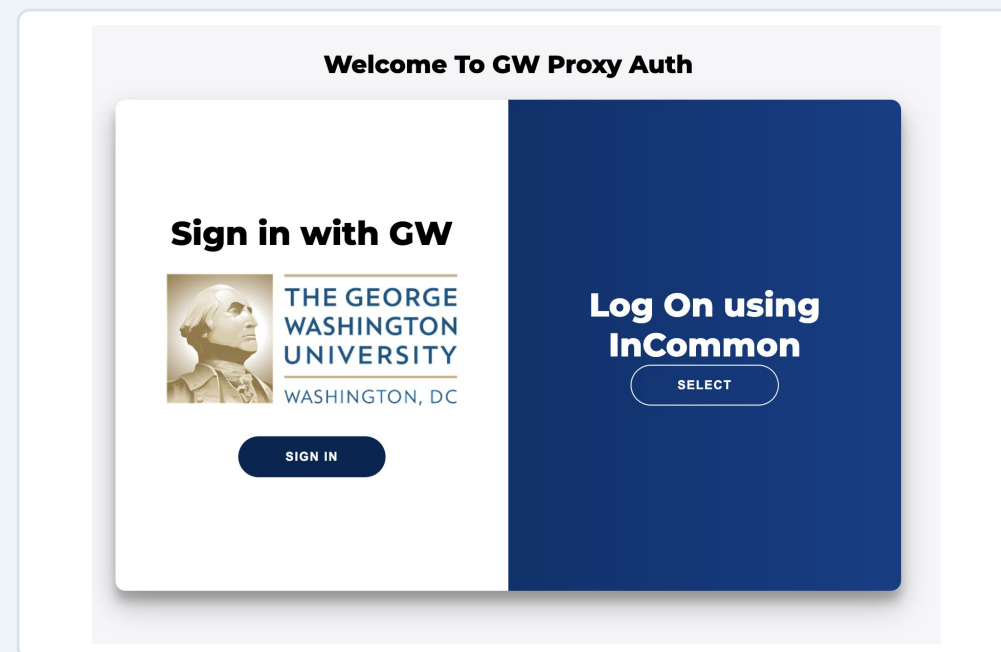
Choose InCommon-Login.



The screenshot shows the login page for The George Washington University Research Gateway. The header features the university's name and location. Below the header, there are two main login options: 'Sign in with your corporate ID' and 'Sign in with your username and password'. The 'Sign in with your corporate ID' option has a button labeled 'InCommon-Login'. The 'Sign in with your username and password' option includes fields for 'Username' and 'Password', a 'Sign in' button, and a link for 'Forgot your password?'. At the bottom, there is a link for 'Need an account? Sign up'.

2 Sign in with GW

A first login may show “requires admin approval.” That is normal; do this step before the session if you can.

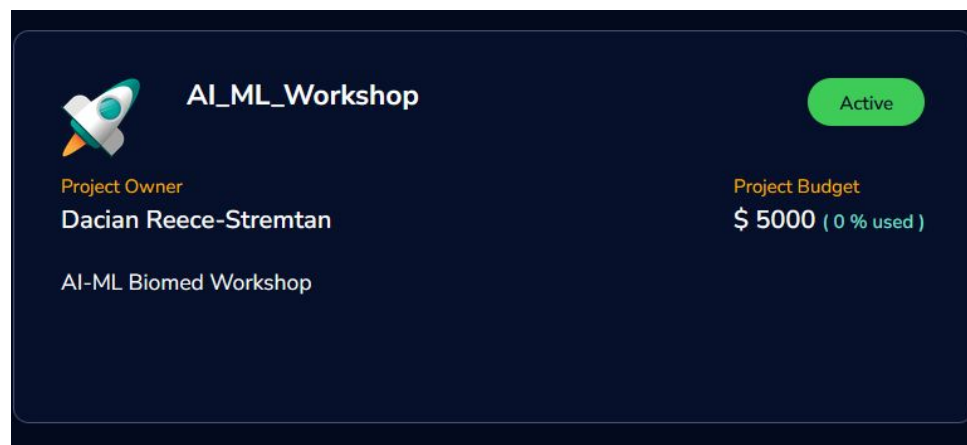


The screenshot shows a 'Welcome To GW Proxy Auth' dialog box. It contains two main sections. The left section, titled 'Sign in with GW', features a portrait of George Washington and the university's name, with a 'SIGN IN' button below. The right section, titled 'Log On using InCommon', has a 'SELECT' button. The background is a solid dark blue color.

Open the project and choose the desktop

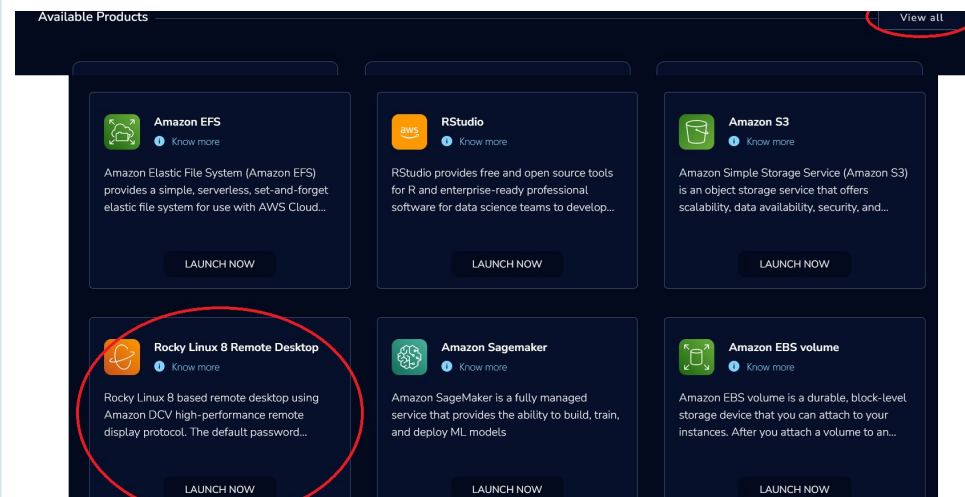
3 Open AI-ML_Workshop

From My Projects, open the AI_ML_Workshop project card.



4 Available Products → Rocky Linux 8 Remote Desktop

Under Available Products, choose Rocky Linux 8 Remote Desktop. You may have to scroll down to see the Rocky Linux 8 Remote Desktop option.



Configure, launch, and sign in

5 Set storage, instance type, password

Set EBSVolumeSize to 80, InstanceType to g5.2xlarge, choose a password, and click Launch Now.

Study Selection
Select one or more studies to mount to your workspace (Maximum of 2)

☒ AI_ML_Workshop(read-write, Internal)

AllowedIpAddress
0.0.0.0/0
The IP address range that can be used to SSH to instance and Connect to DCV

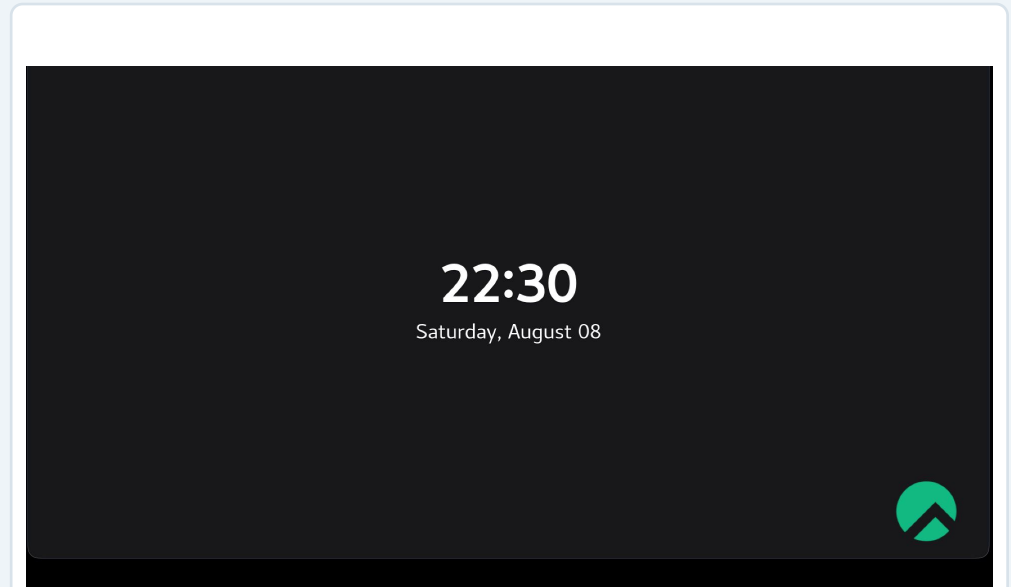
EBSVolumeSize
80
The initial size of the volume (in GBs) EBS will use for storage.

InstanceType
g5.2xlarge
Choose the instance type e.g t3.medium (2vCPU , 2GiB RAM), t3.large (2vCPU, 8GiB RAM), t3.xlarge(4vCPU, 16GiB RAM)

Password
.....
The password must be at least six characters long and include characters from at least two of the following categories- English uppercase letters, English lowercase letters, base-10 digits, and non-alphabetic symbols.

6 Log into the desktop

A new browser tab opens with the desktop. Log in with your password; drag upward if the password box is hidden.



Open Jupyter and start the notebook

Open a terminal on the desktop and run:

```
git clone https://github.com/glenamac/BootcampSquared.git
sudo dnf install python3.12 -y
python3.12 -m venv my_new_env
source my_new_env/bin/activate
pip install notebook
jupyter notebook
```

Then open:

/BootcampSquared/ollama_aws.ipynb

Run the model-pull cells first so the download finishes while you keep working.

Jupyter opens in a browser on the desktop. The virtual environment must be active before you launch it.